

# Combinatorics

## Combinatorics

### MATH 31003-01 - Fall 2026

**Course:** MATH 31003-01

**Semester:** Fall 2026

**Instructor:** Gennady Uraltsev

**Meetings:** Monday, Wednesday, Friday 8:35-9:25 a.m., GEAR 108

Welcome to Combinatorics! We will explore the mathematics of discrete objects: the many ways things can be selected, arranged, connected, and counted. The course covers basic combinatorial techniques, the principle of inclusion-exclusion, generating functions, and recursive relations. Additional topics may include modular arithmetic and applications to encryption, coding theory, graph theory, counting with symmetries, and algebraic structures.

## 0.1 Start here

- Syllabus: how the course works, grading, materials, and what to expect.
- Course calendar: class dates, current topics, holidays, and exams.
- Assignments: written homework and online questions.
- Resources: references and student-support links.
- Lecture notes: notes organized by course unit and lecture.

Please check the course calendar for holidays, University deadlines, and topic updates. Course announcements, grades, and submission links are maintained in Blackboard.